```
#libs
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline

df = pd.read_excel("CoffeeSales.xlsx")
```

```
        transaction_id transaction_date transaction_time
transaction_qty  \
0                    1       2023-01-01         07:06:11
2
1                    2       2023-01-01         07:08:56
2
2                    3       2023-01-01         07:14:04
2
3                    4       2023-01-01         07:20:24
1
4                    5       2023-01-01         07:22:41
2
...                ...              ...              ...
...
149111          149452       2023-06-30         20:18:41
2
149112          149453       2023-06-30         20:25:10
2
149113          149454       2023-06-30         20:31:34
1
149114          149455       2023-06-30         20:57:19
1
149115          149456       2023-06-30         20:57:19
2

        store_id   store_location  product_id  unit_price
product_category  \
0              5  Lower Manhattan          32        3.00
Coffee
1              5  Lower Manhattan          57        3.10
Tea
2              5  Lower Manhattan          59        4.50  Drinking
Chocolate
3              5  Lower Manhattan          22        2.00
Coffee
4              5  Lower Manhattan          57        3.10
Tea
...          ...              ...         ...         ...
...
149111         8   Hell's Kitchen          44        2.50
Tea
```

```
149112         8   Hell's Kitchen          49        3.00
Tea
149113         8   Hell's Kitchen          45        3.00
Tea
149114         8   Hell's Kitchen          40        3.75
Coffee
149115         8   Hell's Kitchen          64        0.80
Flavours

                  product_type               product_detail
0       Gourmet brewed coffee                  Ethiopia Rg
1             Brewed Chai tea     Spicy Eye Opener Chai Lg
2               Hot chocolate            Dark chocolate Lg
3                  Drip coffee  Our Old Time Diner Blend Sm
4             Brewed Chai tea     Spicy Eye Opener Chai Lg
...                       ...                          ...
149111      Brewed herbal tea               Peppermint Rg
149112       Brewed Black tea         English Breakfast Lg
149113      Brewed herbal tea               Peppermint Lg
149114       Barista Espresso                  Cappuccino
149115          Regular syrup               Hazelnut syrup

[149116 rows x 11 columns]

df.head()

   transaction_id transaction_date transaction_time
transaction_qty  \
0               1       2023-01-01         07:06:11                  2

1               2       2023-01-01         07:08:56                  2

2               3       2023-01-01         07:14:04                  2

3               4       2023-01-01         07:20:24                  1

4               5       2023-01-01         07:22:41                  2


   store_id   store_location  product_id  unit_price
product_category  \
0         5  Lower Manhattan          32         3.0
Coffee
1         5  Lower Manhattan          57         3.1
Tea
2         5  Lower Manhattan          59         4.5  Drinking
Chocolate
3         5  Lower Manhattan          22         2.0
Coffee
4         5  Lower Manhattan          57         3.1
```

```
Tea

            product_type               product_detail
0  Gourmet brewed coffee                  Ethiopia Rg
1        Brewed Chai tea     Spicy Eye Opener Chai Lg
2          Hot chocolate            Dark chocolate Lg
3            Drip coffee  Our Old Time Diner Blend Sm
4        Brewed Chai tea     Spicy Eye Opener Chai Lg

df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 149116 entries, 0 to 149115
Data columns (total 11 columns):
 #   Column            Non-Null Count   Dtype
---  ------            --------------   -----
 0   transaction_id    149116 non-null  int64
 1   transaction_date  149116 non-null  datetime64[ns]
 2   transaction_time  149116 non-null  object
 3   transaction_qty   149116 non-null  int64
 4   store_id          149116 non-null  int64
 5   store_location    149116 non-null  object
 6   product_id        149116 non-null  int64
 7   unit_price        149116 non-null  float64
 8   product_category  149116 non-null  object
 9   product_type      149116 non-null  object
 10  product_detail    149116 non-null  object
dtypes: datetime64[ns](1), float64(1), int64(4), object(5)
memory usage: 12.5+ MB

df.describe()

       transaction_id  transaction_qty       store_id
product_id  \
count   149116.000000    149116.000000  149116.000000  149116.000000

mean     74737.371872         1.438276       5.342063      47.918607

std      43153.600016         0.542509       2.074241      17.930020

min          1.000000         1.000000       3.000000       1.000000

25%      37335.750000         1.000000       3.000000      33.000000

50%      74727.500000         1.000000       5.000000      47.000000

75%     112094.250000         2.000000       8.000000      60.000000

max     149456.000000         8.000000       8.000000      87.000000
```

```
          unit_price
count  149116.000000
mean        3.382219
std         2.658723
min         0.800000
25%         2.500000
50%         3.000000
75%         3.750000
max        45.000000
```

```
df.isnull().sum()
```

```
transaction_id      0
transaction_date    0
transaction_time    0
transaction_qty     0
store_id            0
store_location      0
product_id          0
unit_price          0
product_category    0
product_type        0
product_detail      0
dtype: int64
```

```
df.shape
```

```
(149116, 11)
```

```
df["transaction_date"].unique()
```

```
array(['2023-01-01T00:00:00.000000000', '2023-01-
02T00:00:00.000000000',
       '2023-01-03T00:00:00.000000000', '2023-01-
04T00:00:00.000000000',
       '2023-01-05T00:00:00.000000000', '2023-01-
06T00:00:00.000000000',
       '2023-01-07T00:00:00.000000000', '2023-01-
08T00:00:00.000000000',
       '2023-01-09T00:00:00.000000000', '2023-01-
10T00:00:00.000000000',
       '2023-01-11T00:00:00.000000000', '2023-01-
12T00:00:00.000000000',
       '2023-01-13T00:00:00.000000000', '2023-01-
14T00:00:00.000000000',
       '2023-01-15T00:00:00.000000000', '2023-01-
16T00:00:00.000000000',
       '2023-01-17T00:00:00.000000000', '2023-01-
18T00:00:00.000000000',
       '2023-01-19T00:00:00.000000000', '2023-01-
20T00:00:00.000000000',
```

```
       '2023-01-21T00:00:00.000000000', '2023-01-
22T00:00:00.000000000',
       '2023-01-23T00:00:00.000000000', '2023-01-
24T00:00:00.000000000',
       '2023-01-25T00:00:00.000000000', '2023-01-
26T00:00:00.000000000',
       '2023-01-27T00:00:00.000000000', '2023-01-
28T00:00:00.000000000',
       '2023-01-29T00:00:00.000000000', '2023-01-
30T00:00:00.000000000',
       '2023-01-31T00:00:00.000000000', '2023-02-
01T00:00:00.000000000',
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03T00:00:00.000000000',
       '2023-02-04T00:00:00.000000000', '2023-02-
05T00:00:00.000000000',
       '2023-02-06T00:00:00.000000000', '2023-02-
07T00:00:00.000000000',
       '2023-02-08T00:00:00.000000000', '2023-02-
09T00:00:00.000000000',
       '2023-02-10T00:00:00.000000000', '2023-02-
11T00:00:00.000000000',
       '2023-02-12T00:00:00.000000000', '2023-02-
13T00:00:00.000000000',
       '2023-02-14T00:00:00.000000000', '2023-02-
15T00:00:00.000000000',
       '2023-02-16T00:00:00.000000000', '2023-02-
17T00:00:00.000000000',
       '2023-02-18T00:00:00.000000000', '2023-02-
19T00:00:00.000000000',
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23T00:00:00.000000000',
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27T00:00:00.000000000',
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03T00:00:00.000000000',
       '2023-03-04T00:00:00.000000000', '2023-03-
05T00:00:00.000000000',
       '2023-03-06T00:00:00.000000000', '2023-03-
07T00:00:00.000000000',
       '2023-03-08T00:00:00.000000000', '2023-03-
09T00:00:00.000000000',
       '2023-03-10T00:00:00.000000000', '2023-03-
```

```
11T00:00:00.000000000',
       '2023-03-12T00:00:00.000000000', '2023-03-
13T00:00:00.000000000',
       '2023-03-14T00:00:00.000000000', '2023-03-
15T00:00:00.000000000',
       '2023-03-16T00:00:00.000000000', '2023-03-
17T00:00:00.000000000',
       '2023-03-18T00:00:00.000000000', '2023-03-
19T00:00:00.000000000',
       '2023-03-20T00:00:00.000000000', '2023-03-
21T00:00:00.000000000',
       '2023-03-22T00:00:00.000000000', '2023-03-
23T00:00:00.000000000',
       '2023-03-24T00:00:00.000000000', '2023-03-
25T00:00:00.000000000',
       '2023-03-26T00:00:00.000000000', '2023-03-
27T00:00:00.000000000',
       '2023-03-28T00:00:00.000000000', '2023-03-
29T00:00:00.000000000',
       '2023-03-30T00:00:00.000000000', '2023-03-
31T00:00:00.000000000',
       '2023-04-01T00:00:00.000000000', '2023-04-
02T00:00:00.000000000',
       '2023-04-03T00:00:00.000000000', '2023-04-
04T00:00:00.000000000',
       '2023-04-05T00:00:00.000000000', '2023-04-
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       '2023-04-07T00:00:00.000000000', '2023-04-
08T00:00:00.000000000',
       '2023-04-09T00:00:00.000000000', '2023-04-
10T00:00:00.000000000',
       '2023-04-11T00:00:00.000000000', '2023-04-
12T00:00:00.000000000',
       '2023-04-13T00:00:00.000000000', '2023-04-
14T00:00:00.000000000',
       '2023-04-15T00:00:00.000000000', '2023-04-
16T00:00:00.000000000',
       '2023-04-17T00:00:00.000000000', '2023-04-
18T00:00:00.000000000',
       '2023-04-19T00:00:00.000000000', '2023-04-
20T00:00:00.000000000',
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22T00:00:00.000000000',
       '2023-04-23T00:00:00.000000000', '2023-04-
24T00:00:00.000000000',
       '2023-04-25T00:00:00.000000000', '2023-04-
26T00:00:00.000000000',
       '2023-04-27T00:00:00.000000000', '2023-04-
28T00:00:00.000000000',
```

```
       '2023-04-29T00:00:00.000000000', '2023-04-
30T00:00:00.000000000',
       '2023-05-01T00:00:00.000000000', '2023-05-
02T00:00:00.000000000',
       '2023-05-03T00:00:00.000000000', '2023-05-
04T00:00:00.000000000',
       '2023-05-05T00:00:00.000000000', '2023-05-
06T00:00:00.000000000',
       '2023-05-07T00:00:00.000000000', '2023-05-
08T00:00:00.000000000',
       '2023-05-09T00:00:00.000000000', '2023-05-
10T00:00:00.000000000',
       '2023-05-11T00:00:00.000000000', '2023-05-
12T00:00:00.000000000',
       '2023-05-13T00:00:00.000000000', '2023-05-
14T00:00:00.000000000',
       '2023-05-15T00:00:00.000000000', '2023-05-
16T00:00:00.000000000',
       '2023-05-17T00:00:00.000000000', '2023-05-
18T00:00:00.000000000',
       '2023-05-19T00:00:00.000000000', '2023-05-
20T00:00:00.000000000',
       '2023-05-21T00:00:00.000000000', '2023-05-
22T00:00:00.000000000',
       '2023-05-23T00:00:00.000000000', '2023-05-
24T00:00:00.000000000',
       '2023-05-25T00:00:00.000000000', '2023-05-
26T00:00:00.000000000',
       '2023-05-27T00:00:00.000000000', '2023-05-
28T00:00:00.000000000',
       '2023-05-29T00:00:00.000000000', '2023-05-
30T00:00:00.000000000',
       '2023-05-31T00:00:00.000000000', '2023-06-
01T00:00:00.000000000',
       '2023-06-02T00:00:00.000000000', '2023-06-
03T00:00:00.000000000',
       '2023-06-04T00:00:00.000000000', '2023-06-
05T00:00:00.000000000',
       '2023-06-06T00:00:00.000000000', '2023-06-
07T00:00:00.000000000',
       '2023-06-08T00:00:00.000000000', '2023-06-
09T00:00:00.000000000',
       '2023-06-10T00:00:00.000000000', '2023-06-
11T00:00:00.000000000',
       '2023-06-12T00:00:00.000000000', '2023-06-
13T00:00:00.000000000',
       '2023-06-14T00:00:00.000000000', '2023-06-
15T00:00:00.000000000',
       '2023-06-16T00:00:00.000000000', '2023-06-
```

```
17T00:00:00.000000000',
       '2023-06-18T00:00:00.000000000', '2023-06-
19T00:00:00.000000000',
       '2023-06-20T00:00:00.000000000', '2023-06-
21T00:00:00.000000000',
       '2023-06-22T00:00:00.000000000', '2023-06-
23T00:00:00.000000000',
       '2023-06-24T00:00:00.000000000', '2023-06-
25T00:00:00.000000000',
       '2023-06-26T00:00:00.000000000', '2023-06-
27T00:00:00.000000000',
       '2023-06-28T00:00:00.000000000', '2023-06-
29T00:00:00.000000000',
       '2023-06-30T00:00:00.000000000'], dtype='datetime64[ns]')
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 149116 entries, 0 to 149115
Data columns (total 11 columns):
 #   Column            Non-Null Count   Dtype
---  ------            --------------   -----
 0   transaction_id    149116 non-null  int64
 1   transaction_date  149116 non-null  datetime64[ns]
 2   transaction_time  149116 non-null  object
 3   transaction_qty   149116 non-null  int64
 4   store_id          149116 non-null  int64
 5   store_location    149116 non-null  object
 6   product_id        149116 non-null  int64
 7   unit_price        149116 non-null  float64
 8   product_category  149116 non-null  object
 9   product_type      149116 non-null  object
 10  product_detail    149116 non-null  object
dtypes: datetime64[ns](1), float64(1), int64(4), object(5)
memory usage: 12.5+ MB
```

```
df["store_location"].unique()
```

```
array(['Lower Manhattan', "Hell's Kitchen", 'Astoria'], dtype=object)
```

```
df["Total_Amount"] = df["unit_price"] * df["transaction_qty"]
```

```
df.head()
```

```
   transaction_id transaction_date transaction_time
transaction_qty  \
0               1       2023-01-01         07:06:11                2

1               2       2023-01-01         07:08:56                2

2               3       2023-01-01         07:14:04                2
```

```
3               4       2023-01-01         07:20:24                  1

4               5       2023-01-01         07:22:41                  2


   store_id   store_location  product_id  unit_price
product_category  \
0         5  Lower Manhattan          32         3.0
Coffee
1         5  Lower Manhattan          57         3.1
Tea
2         5  Lower Manhattan          59         4.5  Drinking
Chocolate
3         5  Lower Manhattan          22         2.0
Coffee
4         5  Lower Manhattan          57         3.1
Tea

             product_type                product_detail  Total_Amount
0  Gourmet brewed coffee                   Ethiopia Rg           6.0
1        Brewed Chai tea     Spicy Eye Opener Chai Lg           6.2
2          Hot chocolate            Dark chocolate Lg           9.0
3            Drip coffee  Our Old Time Diner Blend Sm           2.0
4        Brewed Chai tea     Spicy Eye Opener Chai Lg           6.2
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 149116 entries, 0 to 149115
Data columns (total 12 columns):
 #   Column            Non-Null Count   Dtype
---  ------            --------------   -----
 0   transaction_id    149116 non-null  int64
 1   transaction_date  149116 non-null  datetime64[ns]
 2   transaction_time  149116 non-null  object
 3   transaction_qty   149116 non-null  int64
 4   store_id          149116 non-null  int64
 5   store_location    149116 non-null  object
 6   product_id        149116 non-null  int64
 7   unit_price        149116 non-null  float64
 8   product_category  149116 non-null  object
 9   product_type      149116 non-null  object
 10  product_detail    149116 non-null  object
 11  Total_Amount      149116 non-null  float64
dtypes: datetime64[ns](1), float64(2), int64(4), object(5)
memory usage: 13.7+ MB
```

```
df["product_category"].unique()
```

```
array(['Coffee', 'Tea', 'Drinking Chocolate', 'Bakery', 'Flavours',
       'Loose Tea', 'Coffee beans', 'Packaged Chocolate', 'Branded'],
      dtype=object)
```

```
df["product_type"].unique()
```

```
array(['Gourmet brewed coffee', 'Brewed Chai tea', 'Hot chocolate',
       'Drip coffee', 'Scone', 'Barista Espresso', 'Brewed Black tea',
       'Brewed Green tea', 'Brewed herbal tea', 'Biscotti', 'Pastry',
       'Organic brewed coffee', 'Premium brewed coffee', 'Regular
syrup',
       'Herbal tea', 'Gourmet Beans', 'Organic Beans', 'Sugar free
syrup',
       'Drinking Chocolate', 'Premium Beans', 'Chai tea', 'Green
beans',
       'Espresso Beans', 'Green tea', 'Organic Chocolate',
'Housewares',
       'Black tea', 'House blend Beans', 'Clothing'], dtype=object)
```

```
df["store_id"].unique()
```

```
array([5, 8, 3], dtype=int64)
```

```
total_transaction = df["transaction_id"].count()
total_quantiy = df["transaction_qty"].sum()
```

```
total_quantiy
```

```
214470
```

```
amount_over_the_month.columns
```

```
Index(['sum', 'count'], dtype='object')
```

```
df["Month"] = df["transaction_date"].dt.month
```

```
df.head()
```

```
   transaction_id transaction_date transaction_time
transaction_qty  \
0               1       2023-01-01         07:06:11                 2

1               2       2023-01-01         07:08:56                 2

2               3       2023-01-01         07:14:04                 2

3               4       2023-01-01         07:20:24                 1

4               5       2023-01-01         07:22:41                 2


   store_id   store_location  product_id  unit_price
```

```
product_category  \
0         5  Lower Manhattan          32         3.0
Coffee
1         5  Lower Manhattan          57         3.1
Tea
2         5  Lower Manhattan          59         4.5  Drinking
Chocolate
3         5  Lower Manhattan          22         2.0
Coffee
4         5  Lower Manhattan          57         3.1
Tea

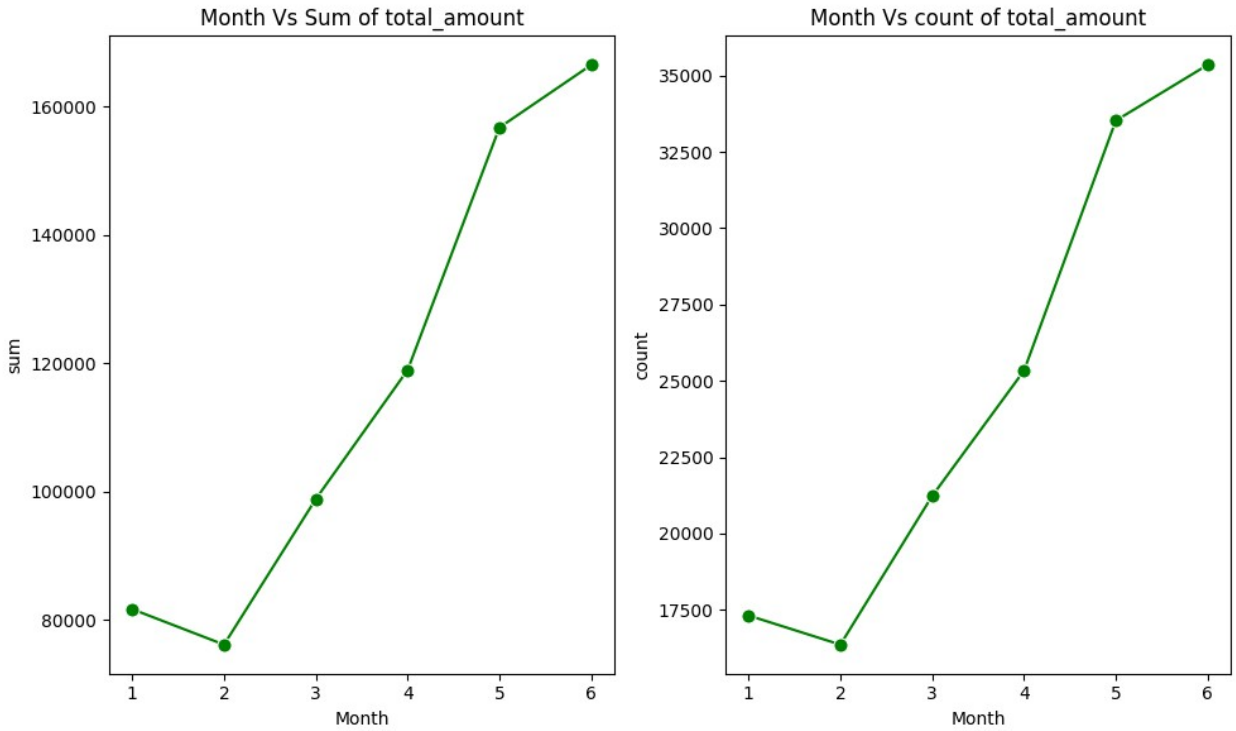
             product_type                product_detail  Total_Amount
Month
0  Gourmet brewed coffee                   Ethiopia Rg           6.0
1
1        Brewed Chai tea     Spicy Eye Opener Chai Lg           6.2
1
2          Hot chocolate            Dark chocolate Lg           9.0
1
3            Drip coffee  Our Old Time Diner Blend Sm           2.0
1
4        Brewed Chai tea     Spicy Eye Opener Chai Lg           6.2
1
```

```
amount_over_the_month = df.groupby("Month")
["Total_Amount"].aggregate(["sum","count"])

plt.figure(figsize=(10,6))
plt.subplot(1,2,1)
sns.lineplot(data=amount_over_the_month,x=amount_over_the_month.index,
y="sum",color="green",marker="o",markersize=8)
plt.title("Month Vs Sum of total_amount")

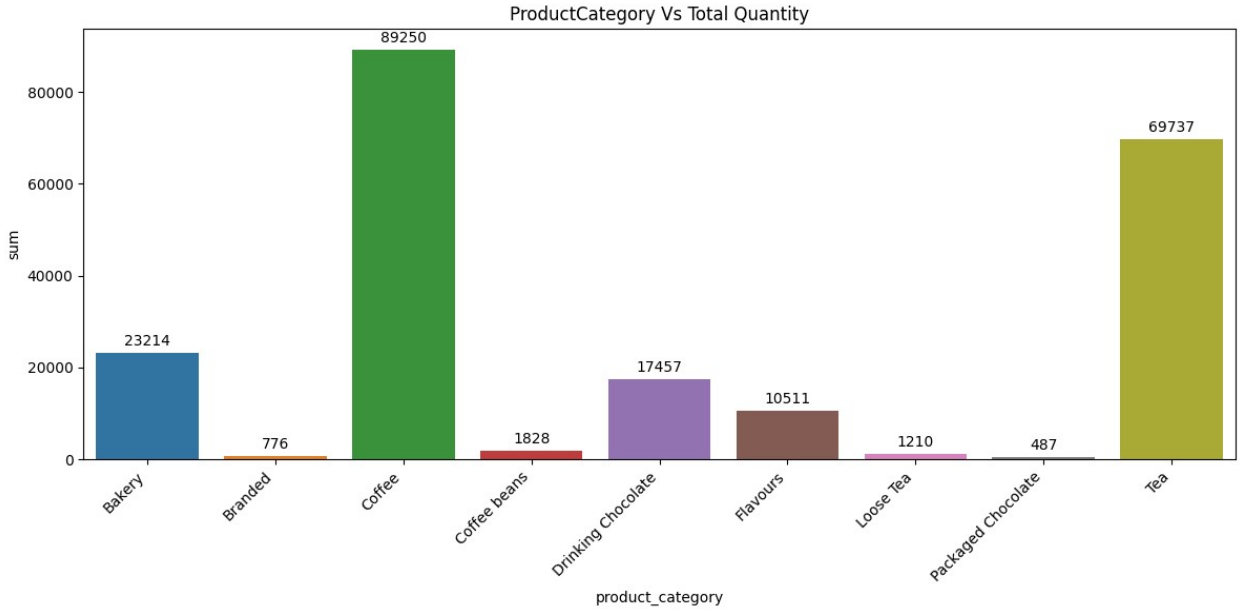
plt.subplot(1,2,2)
sns.lineplot(data=amount_over_the_month,x=amount_over_the_month.index,
y="count",color="green",marker="o",markersize=8)
plt.title("Month Vs count of total_amount")

plt.tight_layout()
plt.show()
```

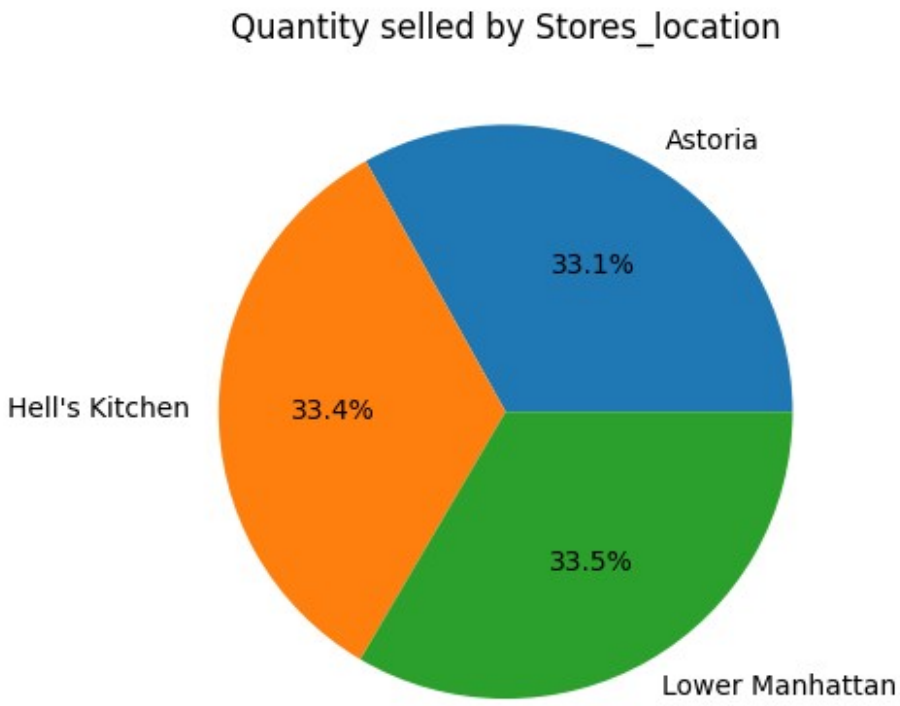



```
product_category_by_quantity = df.groupby("product_category")
["transaction_qty"].aggregate(["sum"])
store_location_sell_quantity = df.groupby("store_location")
["transaction_qty"].aggregate(["sum"])

plt.figure(figsize=(12,6))
ax =
sns.barplot(data=product_category_by_quantity,x=product_category_by_qu
antity.index,y="sum")
for container in ax.containers:
    ax.bar_label(container,fmt="%.0f",label_type="edge",padding=3)
plt.title("ProductCategory Vs Total Quantity")
plt.xticks(rotation=45,ha="right")
plt.tight_layout()
```




```
plt.pie(x=store_location_sell_quantity["sum"],labels=store_location_se
ll_quantity.index,autopct="%.1f%%")
plt.title("Quantity selled by Stores_location")
plt.show()
```




```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 149116 entries, 0 to 149115
Data columns (total 13 columns):
 #   Column            Non-Null Count   Dtype         
---  ------            --------------   -----         
 0   transaction_id    149116 non-null  int64         
 1   transaction_date  149116 non-null  datetime64[ns]
 2   transaction_time  149116 non-null  object        
 3   transaction_qty   149116 non-null  int64         
 4   store_id          149116 non-null  int64         
 5   store_location    149116 non-null  object        
 6   product_id        149116 non-null  int64         
 7   unit_price        149116 non-null  float64       
 8   product_category  149116 non-null  object        
 9   product_type      149116 non-null  object        
 10  product_detail    149116 non-null  object        
 11  Total_Amount      149116 non-null  float64       
 12  Month             149116 non-null  int64         
dtypes: datetime64[ns](1), float64(2), int64(5), object(5)
memory usage: 14.8+ MB
```

FinalReport - Coffee Sales 2023

- Total Transaction - 1,49,116
- Total Quantity - 2,14,470
- sum of TotalAmount sell over month is increased
- Count of TotalAmount sell over month is increased
- ProductCategory by selled quantity - Coffee and Tea is secured more sells
- quantity selled based on location - Both 3 locations approximately selled equal quantity over the month